NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

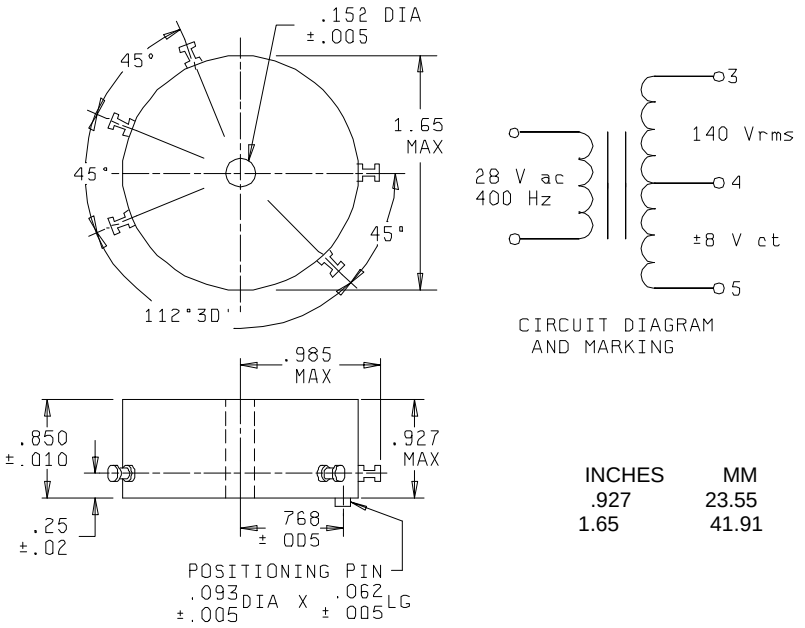
MIL-PRF-27/33
6 August 1975

PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-UP, TF5S03ZZ319

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.927	23.55
1.65	41.91

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Marking shall be on the side of the case.
4. Terminal angular position tolerance is ±1°.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS (When numbers in parentheses, i e , (1-2) are used, they indicate the winding and the extreme terminals of the winding)

Electrical ratings

Primary voltage,
(1-2) 28 volts rms, 400 Hz ±5 percent

Secondary voltage,
(3-4) 70 volts rms, ±4 volts rms
(3-5) 140 volts rms, ±8 volts rms centertap
(4-5 70 volts rms, ±4 volts rms

Working voltage,
(1-2) 200 volts maximum
(3-5) 200 volts maximum

Design and construction

Dimensions and configuration See figure 1

Duty cycle Continuous

Case Encapsulated

Material Diallyl-phthalate

Terminals Turret, solder type, brass, hot tinned dip
Terminal height 0 156 inch +0 005

Weight 4 ozs

Operating temperature range -55°C to 130°C

Terminal strength Method 211, MIL-STD-202, test condition A, 2 pounds

Dielectric withstanding voltage (at sea level) 560 volts rms

Electrical characteristics

No load With 28 volts rms, 400 Hz across (1-2),

Voltage across (3-4) 70 volts rms ±4 volts, and the voltage across (4-5) 70 volts rms ±4 volts

Rated load With 28 volts, 400 Hz across (1-2), the voltage across (3-5) shall be 140 volts rms centertap, ±8 volts

DC resistance,
(1-2) 16 5 ohms ±20 percent
(3-4) 97 ohms ±20 percent
(4-5) 112 ohms ±20 percent

Polarity Additive with 2 and 3 connected

Temperature rise 30°C maximum

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Marking location See figure 1

Part number M27/33-01

Custodians
  Army - EL
  Navy - EC
  Air Force - 85

Review activities
  Army - EL, MI, SG
  Navy - EC, SH, OS
  Air Force - 11, 17, 80
  DSA - ES

User activities
  Army - MU, WC, SG, ME
  Navy - MC, AS
  Air Force - 19

Preparing activity
  Army - EL

Agent
  DSA - ES

(Project 5950-0490)

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION MIL-T-27/33 TRANSFORMER, POWER, STEP-UP, TF5S03ZZ319

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes" in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity Optional)*

DATE

DD FORM 1426 REPLACES DD FORM 1426, 1 JAN 64, WHICH MAY BE USED

ESC-FM 1068-68

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